

E-Commerce Order Profit Prediction using Machine Learning

Insights

This project focuses on building an end-to-end Machine Learning Regression model to predict expected profit using historical Superstore sales data. The project includes preprocessing, exploratory data analysis, feature engineering, model training, hyperparameter tuning, and deployment using Streamlit.

1. Business Problem

Many retail businesses generate high sales but still suffer low profits due to excessive discounts, operational costs, and poor-performing categories or regions. This project helps businesses predict future profit and identify risky orders.

2. Business Objectives

- Predict expected order profit.
- Identify potential loss-making orders.
- Understand factors affecting profitability.
- Optimize discount strategy.
- Build a deployable prediction system.

3. Dataset Information

The Superstore dataset contains 9,924 rows and 21 columns. It includes order details, sales, discounts, quantity, category, region, shipping information, and profit values.

4. Project Workflow

1. Data Loading
2. Data Preprocessing
3. Exploratory Data Analysis
4. Feature Engineering
5. Encoding & Transformation
6. Model Building
7. Hyperparameter Tuning
8. Cross Validation
9. Streamlit Deployment

5. EDA Insights

- Technology category generated highest profit.
- West region achieved highest profitability.
- High discounts reduced overall profit.
- Some high-sales orders generated losses.
- Profit distribution was slightly skewed.
- Furniture category showed lower profit margins.

6. Feature Engineering

New features created: Delivery Days, Order Month, Order Year, Discount Amount, and Unit Price. These features improved model understanding of business operations.

7. Model Comparison

Model	R ² Score
Random Forest Regressor	0.8347
Decision Tree Regressor	0.6921
Linear Regression	0.5573
KNN Regressor	0.4827

8. Final Evaluation Metrics

- Final Test R² Score: **0.8347**
- Cross Validation Mean Score: **0.8348**
- MAE: **5.56**
- RMSE: **11.78**

9. Deployment

The final model was deployed using Streamlit. Users can enter order details and instantly receive profit or loss predictions.

10. Conclusion

This project successfully transformed historical retail data into a predictive machine learning system. The final Random Forest model achieved approximately **83.4% R²** accuracy and provided meaningful business insights related to discount strategy, regional performance, and category profitability.